

Kunal Avghade

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Summary

Full Stack Software Engineer with **2.5+ years** of experience building high-scale distributed systems and cloud-native architectures. Specialist in Java/Spring Boot and Python/FastAPI, with a proven track record of delivering fault-tolerant platforms and managing **terabyte-scale** data migrations. AWS Certified Solutions Architect with strong algorithmic foundations (**Top 7% LeetCode**).

Technical Skills

Languages: Java, Python, C++, JavaScript, TypeScript, SQL, HTML/CSS
Backend: Spring Boot, NestJS, FastAPI, REST, gRPC, Temporal, Kafka, RabbitMQ, Hibernate
Frontend: React.js, Next.js, Redux, JavaScript
Cloud/DevOps: AWS (EC2, S3, Lambda, SQS), GCP, Docker, Kubernetes, Jenkins, ArgoCD
Databases: PostgreSQL, Redis, MongoDB, Vector Databases (Pinecone, Milvus)

Professional Experience

Associate Software Engineer | Calfus Inc.

Jul 2023 – Present

Full Stack Development & Cloud-Native Engineering

Pune, India

• Aetherion — AI Agent SDK & Low-Code Platform

Tech Stack: Python, FastAPI, Temporal, Pydantic, PostgreSQL, Docker, AWS

- Developed a production-grade **Python SDK** for seamless integration with the Aetherion AI ecosystem, allowing external developers to build and deploy custom agents with 40% less boilerplate code.
- Architected the core backend for a **Low-Code Agent Builder**, utilizing **Temporal** to orchestrate complex, long-running agent workflows with native fault tolerance, retries, and state persistence.
- Designed extensible prompt management and tool-calling interfaces, facilitating the rapid onboarding of new LLM providers and custom toolsets.

• NetApp Data Migrator — Enterprise Storage Solution

Tech Stack: Java, Spring Boot, NestJS, gRPC, Redis, RabbitMQ, Keycloak, PostgreSQL

- Engineered a high-performance migration engine capable of transferring **multi-terabyte** datasets between NAS (NFS/SMB) systems with 99.9% data integrity across hybrid-cloud environments.
- Optimized discovery agents using **gRPC** for low-latency communication and **multi-threaded processing**, achieving a 3x increase in file-scanning speed compared to legacy implementations.
- Implemented distributed locking via **Redis** to manage concurrent migration tasks across worker nodes.

• Asurint — Background Verification Microservices

Tech Stack: Java, Spring Boot, GCP, Kafka, ArgoCD, Jenkins, PostgreSQL

- Built and maintained scalable microservices on **GCP** to automate background check workflows, processing millions of requests for criminal record indexing and education verification.
- Orchestrated asynchronous inter-service communication using **Kafka**, decoupling heavy report generation logic from real-time data ingestion.

Key Project

Vision-Based Intelligent Recipe Recommender

Tech Stack: Python, Django, YOLOv8, OpenCV, React, PostgreSQL, JWT

Jan 2023 – May 2023

- Engineered a computer vision application that identifies food ingredients from user images using a custom-trained **YOLOv8** model with **92% Mean Average Precision (mAP)**.
- Developed a **Django** REST API to map detected objects to a database of 5000+ recipes using optimized SQL queries.
- Implemented secure user authentication and ingredient history tracking using **JWT** and a React frontend.

Achievements & Certifications

- LeetCode:** Ranked in the **Top 7%** globally (Max Rating: 1900+); Solved **1000+** DSA problems. | Profile
- Drive Dynamo Award:** Won **3 years consecutively** (2023, 2024, 2025) for exceptional technical delivery and leading technology adaptation at Calfus Inc.
- AWS Certified Solutions Architect – Associate & AWS Cloud Practitioner.**
- IBM AI Engineering Professional & Stanford Machine Learning Specialization.**

Education

Vishwakarma Institute of Information Technology

B.Tech in Electronics & Telecommunication (Software Systems Focus)

Pune, India

2020 – 2023